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D205

Performance Assessment

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Insulin plays an important role in how sugar is stored in our bodies. When your body has trouble producing enough insulin and correctly regulating your blood sugar levels, you have diabetes. There are two types. Type I: a person is born with their immune system attacking the cells that are supposed to produce insulin. Type II: develops over time especially in people who are obese and inactive, making them insulin-resistant (Felman & Patel (PharmD, BCPS), 2023).

According to the American Diabetes Association, many studies have shown that there might be an association between Type II diabetes and low vitamin D levels. This SQL query will look at the number of people in the East Coast states that have low vitamin D levels and diabetes (Talaie, Mohamadi, & Adgi, 2013).

Vitamin D levels are split into three categories (measured in nanograms per millilitre) (Ringe & Kipshoven, 2012):

Optimal: level above 30 ng/ml

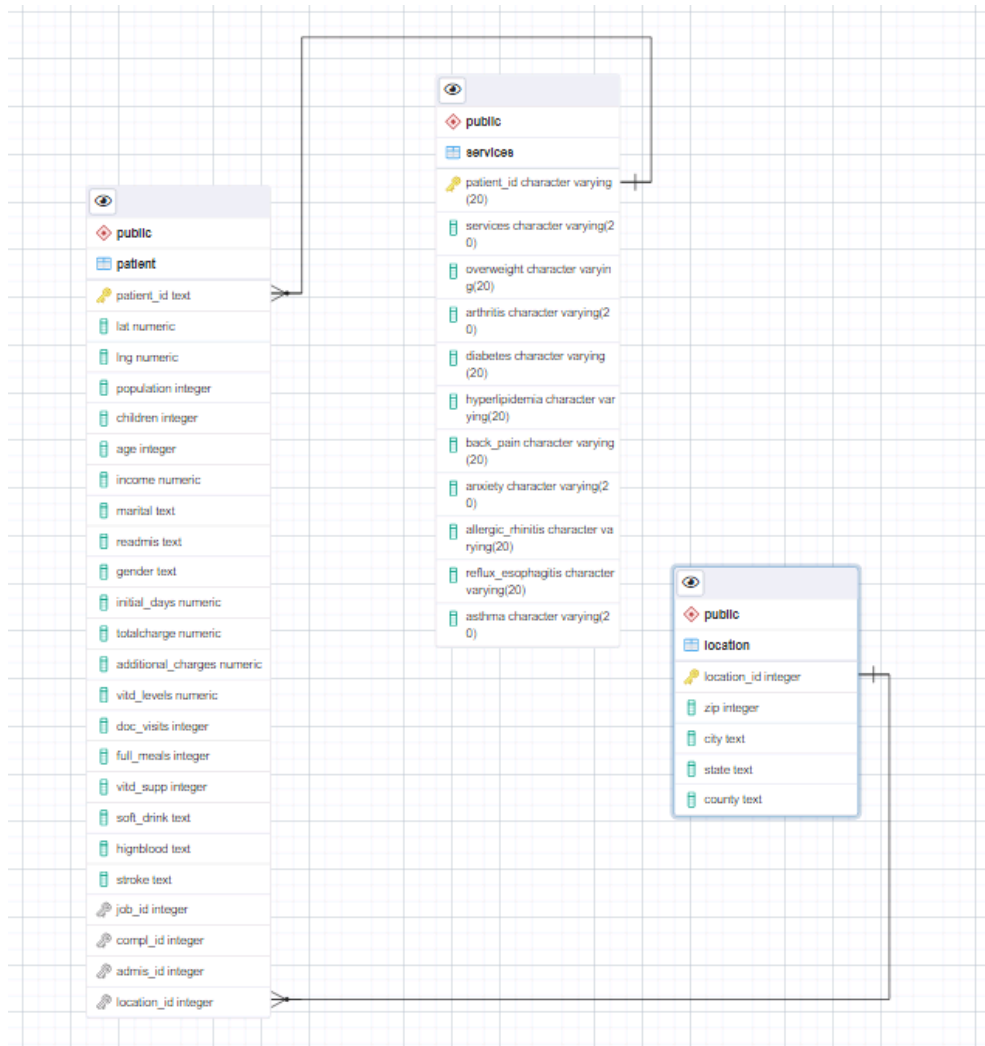
Suboptimal: level between 20-29 ng/ml

Deficient: level below 20 ng/ml

A. How many people from the East Coast States fall into one of the three categories of vitamin D levels, and do they have diabetes?

1. From the *location* table: location_id (integer), state (text)
 From the *patient* table: patient_id (text), vitd_levels (integer),
 From the *services* table: patient_id (varchar(20)), diabetes (varchar(20))

B.



1.

DROP TABLE services;

-- The drop table command drops previous versions of the *services* table

```
CREATE TABLE services(  
patient_id VARCHAR(20) NOT NULL PRIMARY KEY,  
services VARCHAR(20),  
overweight VARCHAR(20),  
arthritis VARCHAR(20),  
diabetes VARCHAR(20),  
hyperlipidemia VARCHAR(20),  
back_pain VARCHAR(20),  
anxiety VARCHAR(20),  
allergic_rhinitis VARCHAR(20),  
reflux_esophagitis VARCHAR(20),  
asthma VARCHAR(20)  
);
```

-- The create table command creates the *services* table with its 11 columns. The columns are all of type varchar with a length of 20 since the external CSV only holds short text inputs.

-- The patient_id column is the primary key because it holds unique values, and it cannot be null.

2.

```
"\copy public.services (patient_id, services, overweight, arthritis, diabetes, hyperlipidemia, back_pain,  
anxiety, allergic_rhinitis, reflux_esophagitis, asthma) FROM 'C:/LabFiles/Medical/MSERVI~1.CSV' CSV  
HEADER QUOTE '\"' ESCAPE '\"';"
```

C.

```
SELECT l.state,
```

-- A case statement is needed to split the vitamin D levels into the 3 categories.

```
    CASE  
        WHEN p.vitd_levels > 30 THEN 'Optimal'  
        WHEN p.vitd_levels < 20 THEN 'Deficient'  
        ELSE 'Suboptimal' END AS vit_d_cat,  
    COUNT(p.patient_id),  
    s.diabetes
```

```
FROM patient AS p  
LEFT JOIN location AS l  
    USING (location_id)  
LEFT JOIN services AS s  
    USING (patient_id)
```

-- Only the East Coast states will be included

```
WHERE state IN ('CT', 'DE', 'FL', 'GA', 'ME', 'MD', 'MA', 'NH', 'NJ', 'NY', 'NC', 'RI', 'SC', 'VA', 'PA', 'VT', 'WV')  
GROUP BY l.state, vit_d_cat, s.diabetes  
ORDER BY l.state, vit_d_cat;
```

1. LFerrusi_D205.csv

D. The add-on file should be acquired and refreshed monthly.

1. Since this information can be used for research purposes, but not for immediate care of patients, it does not need to be updated on a more regular basis.

E. <https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=430b58fb-1d2c-4618-8075-b0940153445c>

References

Felman, A., & Patel (PharmD, BCPS), A. (2023, May 15). *An overview of insulin*. Retrieved from Medical News Today: <https://www.medicalnewstoday.com/articles/323760>

Ringe, J. D., & Kipshoven, C. (2012, January 1). *Dermato Endocrinology*. Retrieved from National Library of Medicine: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3408996/#:~:text=Currently%20a%20level%20of%20at,ng%2Fml%20as%20severe%20deficiency>.

Talaei, A., Mohamadi, M., & Adgi, Z. (2013, February 26). *The effect of Vitamin D on insulin resistance patients with type 2 diabetes*. Retrieved from DMS Journal: <https://doi.org/10.1186/1758-5996-5-8>